Inching past Probabilies or Podcast Survey Results

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Probablity is not (yet) my jam. Left to my own devices, I would simply take tutorial after tutorial, class after class, and read book after book. Thankfully, I have an amazing friend who told me to start going to Meetups. So I attended a tutorial hosted by Raja Iqbal, and learned one key lesson: Show your work.

The work, in this case, is demonstrating how I pulled survey results from Google, cleaned it, and started visualizing it. While this project is not directly related to developing machine learning tools for risk classification, it has given me a better undresetanding of the process of loading, cleaning, and visualizing real world (and useful) data.

The Survey

In preperation for launching a crowd-fund solution, my podcast crew and I decided to ask our fans what we should do to improve our show. So we put a form together and asked the following questions:

```
##  [1] "Email Address"
##  [2] "Why do you listen to the show? Pick all the reasons that apply."
##  [3] "What do you like us to talk about? Pick all the topics you like"
##  [4] "How did you hear about All Comics Considered?"
##  [5] "How do you listen to All Comics Considered?"
##  [6] "Is there another podcast platform that you'd prefer?"
##  [7] "What other podcasts do you listen to?"
##  [8] "Do you listen to podcasts on YouTube?"
##  [9] "What other podcast or comic book based communities do you participate in?"
## [10] "What makes you want to turn All Comics Considered off?"
## [11] "How would you describe All Comics Considered to someone else?"
## [12] "Do you suggest All Comics Considered to friends?  Why or why not?"
## [13] "Have you read the All Comics Considered blog in the past?"
## [14] "Would you be interested in more written content?"
## [15] "How much do you like the Pullbox Feature?"
## [16] "What would improve the Pullbox Feature?"
## [17] "How much do you enjoy our deep dive features into specific comics, properties, and issues?"
## [18] "How much do you enjoy our shorter, more comedic segments?"
## [19] "How do you feel about the length of the show?"
## [20] "What format changes might you be interested in seeing come to the show?"
```

26 people responded to these question. Google Forms has some simple analytics built into the platform, and given the number of responses, a good deal of the insights we gathered from our survey was pulled from that front page. But pulling insights from a form was not how I wanted to access the responses. So I took the opportunity this survey created to see how well I could use R to pull the data.

Making it More Complicated

The first step was determining which package I should use to pull the responses. It took a grand total of two minutes to decide on the package: `googlesheets`. This package, by Jennifer Bryan and Joanna Zhao, has

simple, clear, and consistent syntax. Once my Google credentials[1] were authenticated, pulling data from the sheets was a breeze. The function `gs_ls()` pulled a list of my sheets into a tibble, which made finding the key to the survey responses as simple as: `gs_ls %>% filter(str_detect(sheet_title, "All Comics Considered: 2018")) %>% select(sheet_key) %>% as.character()`.[2]

Armed with the key, I pulled the survey resultls into a single tibble:

```
register_survey <- gs_key(survey_key)
survey_full <- gs_read(register_survey)
survey_anon <- survey_full %>% select(-2)
```

This could have been collapsed into one line of code, which I write in footnote 2. Both methods produce the following table

Overwhelming Table

Timestamp

Why do you listen to the show? Pick all the reasons that apply.

What do you like us to talk about? Pick all the topics you like

How did you hear about All Comics Considered?

How do you listen to All Comics Considered?

Is there another podcast platform that you'd prefer?

What other podcasts do you listen to?

Do you listen to podcasts on YouTube?

What other podcast or comic book based communities do you participate in?

What makes you want to turn All Comics Considered off?

How would you describe All Comics Considered to someone else?

Do you suggest All Comics Considered to friends? Why or why not?

Have you read the All Comics Considered blog in the past?

Would you be interested in more written content?

How much do you like the Pullbox Feature?

What would improve the Pullbox Feature?

How much do you enjoy our deep dive features into specific comics, properties, and issues?

How much do you enjoy our shorter, more comedic segments?

How do you feel about the length of the show?

What format changes might you be interested in seeing come to the show?

3/3/2018 20:14:32

Timely and relevant content., Social relevance., The selection of books talked about.

Comics TV, Comic Movies, Historical comic titles, Social Justice Issues in Comics

[1] Jennifer's vingette explains authentication, where tokens live, and everything you wanted to know about how the package works.

[2] I was not, originally, so clever with my code. First I stored my sheets in a tibble like so, `sheets <- gs_ls()`, then I may have cut and pasted instead of using `stringr` and `as.character()`. Writing this article now, I realize it could have been as easy as `gs_ls() %>% filter(str_detect(sheet_title, "All Comics Considered: 2018")) %>% select(sheet_key) %>% as.character() %>% gs_key() %>% gs_read()`.

facebook

Google Play

itunes

NA

Yes

NA

NA

itchy

sure

Yes

Yes

5

NA

5

5

3

Full episodes about a single topic

3/3/2018 20:30:15

Timely and relevant content., Banter between hosts., Social relevance., The selection of books talked about., Interviews., The best show I have found for balancing love of the medium with a commitment to keeping its feet to the fire when it comes to representation.

Comics TV, Comic Movies, Ongoing comic titles, Historical comic titles, Social Justice Issues in Comics, Ourselves, I think you have the balance right now. Being album to compare the scenarios in different comics to aspects of your own life makes your analysis more grounded.

Websearch. Serendipity.

iTunes

I use Otto occasionally. Mainly iTunes

Where to start: 95BFM; Adventure: A dungeons and dragons podcast; All in the mind; All songs considered; RNZ At the movies; Book Riot; Can He Do That?; Talking Comic; Freakonomics Radio; Fresh Air; Gone by Lunchtime; Larry Wilmore; Mental Illness Happy Hour; NPR Politics Podcast; On One with Angela Rye; Pod Save America; Pod Save the World; Pop Culture Happy Hour; Race Invaders; Still ProcessingThe Ezra Klein Show; Up, Up and Away; WTF with Marc Maron; Women of Marvel

No

I listen to a lot of Podcasts, but this is the only one I really interact with on Facebook etc.

Nothing.

Smart, witty, great coverage of the intersection of politics and comics.

I've actually included the podcast in a 'recommended' list on a radio spot I used to do for my local student radio station.

NA

Yes

5

I like the idea of regular Deep Cuts (great story arcs from the past)

5

4

2

Full episodes about a single topic

This table violates one of the most critical lessons I learned in grad school: The values are not atomic. In other words, the answer to the question `names(survey_anon[2])` have more than one answer per respondant:

```
##  [1] "Timely and relevant content., Social relevance., The selection of books talked about."
##  [2] "Timely and relevant content., Banter between hosts., Social relevance., The selection of books
##  [3] "Banter between hosts., Social relevance., The selection of books talked about., Interviews."
##  [4] "Banter between hosts., Social relevance., Interviews."
##  [5] "Banter between hosts., Social relevance., The selection of books talked about."
##  [6] "Banter between hosts., The selection of books talked about."
##  [7] "Timely and relevant content., Banter between hosts., Social relevance."
##  [8] "Banter between hosts., Social relevance."
##  [9] "Timely and relevant content., Banter between hosts., Social relevance., Interviews."
## [10] "Timely and relevant content., Banter between hosts."
## [11] "Timely and relevant content., Banter between hosts., Social relevance., Interviews., Perspecti
## [12] "Banter between hosts., Social relevance."
## [13] "Timely and relevant content., Banter between hosts., Social relevance., The selection of books
## [14] "Timely and relevant content., Social relevance."
## [15] "Banter between hosts., Social relevance., Interviews."
## [16] "Banter between hosts., The selection of books talked about., Interviews."
## [17] "The selection of books talked about., Interviews."
## [18] "Timely and relevant content., Banter between hosts."
## [19] "Timely and relevant content., Banter between hosts., Social relevance., The selection of books
## [20] "Banter between hosts., Social relevance., The selection of books talked about., To privately la
## [21] "Timely and relevant content., Banter between hosts., Interviews."
## [22] "Timely and relevant content., Social relevance."
## [23] "Timely and relevant content., The selection of books talked about."
## [24] "Timely and relevant content., Social relevance."
## [25] "The selection of books talked about."
## [26] "Timely and relevant content., Social relevance., The selection of books talked about."
```

In addition to the multiple answers per item, we also have a few likard scale questions to capture feelings about the show. To be blunt, our data are a hot mess. Google makes it easy to get basic answers, but if I wanted to use anything other than their tools, there would be a lot of work to clean it up. Which, to be fair, is exactly what I want to do.

Cleaning

With Data Camp, one usually has a way to check their work. With our podcast survey, there is a guide in the form of Google's visiualizations, but nothing about intermediate steps of cleaning or tidying. In order to get any insights from this data, I was going to have tidy it myself.

My first step was to identify any pattern in the responses to pull out individual answers. That was fairly easy for most of the questions, as the default character in Forms is `","`.

```r
a2_wordcloud <- a2  %>% unnest_tokens(word, answers) %>% anti_join(stop_words) %>%
  count(word) %>% filter(n > 5) %>%
  with(wordcloud(word, n, colors = spectral, random.order = FALSE, random.color = FALSE))
```